

Notes: Nakamura and Steinsson (QJE, 2018)

High-Frequency Identification of Monetary Non-Neutrality: The Information Effect

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1 Big picture

- **Question.** What can high-frequency financial-market reactions around FOMC announcements tell us about monetary non-neutrality, and do those reactions identify pure monetary-policy shocks or also Federal Reserve information effects?
- **Empirical strategy.** The paper uses unexpected changes in interest rates in a narrow 30-minute window around scheduled FOMC announcements. The identifying idea is that market movements inside this window are driven by new information released at the announcement.
- **Main empirical fact 1.** A surprise monetary tightening raises nominal Treasury yields and real yields from TIPS roughly one-for-one at horizons several years out in the yield curve.
- **Main empirical fact 2.** Expected inflation, measured using breakeven inflation, responds little at short horizons and only modestly at longer horizons.
- **Main empirical fact 3.** Expected output growth rises after a surprise tightening. This is the opposite of the prediction from a conventional pure monetary-tightening interpretation.
- **Interpretation.** FOMC announcements move beliefs not only about current and future policy rates, but also about economic fundamentals. A rate hike can be read by markets as good news about the Fed's assessment of the economy.
- **Structural model.** The authors estimate a New Keynesian model with internal habits, a backward-looking Phillips curve term, persistent forward-guidance shocks, and a Fed information effect.
- **Key parameter.** The information parameter ψ measures the fraction of the monetary announcement that shifts private-sector beliefs about the natural rate of interest and natural output. The baseline estimate is around 0.68.
- **Main conclusion.** High-frequency monetary shocks are not pure policy shocks. They contain an important information component, and ignoring that component can mislead inference about monetary non-neutrality and the slope of the Phillips curve.
- **Policy implication.** Monetary announcements can be powerful because they change beliefs about fundamentals. But this also means that the Fed may partly “fight against itself” when a tightening communicates good news or an easing communicates bad news.

2 Incremental contribution of the paper

2.1 Contribution relative to high-frequency monetary identification

- Earlier high-frequency studies use narrow windows around FOMC announcements to isolate monetary news from normal macroeconomic endogeneity.
- This paper pushes the approach beyond nominal interest rates. It uses real yields from TIPS, breakeven inflation, Blue Chip forecast revisions, and stock prices to identify the information content of announcements.
- The incremental contribution is to show that high-frequency monetary surprises do not necessarily have the interpretation usually assigned to them: they may mix policy news with Fed information about fundamentals.

Interpretation: The paper does not reject high-frequency identification. It refines it. Narrow-window shocks are cleanly tied to FOMC announcements, but the content of the announcement is richer than a pure exogenous policy-rate disturbance.

2.2 Contribution relative to VAR and narrative monetary-shock studies

- VAR and narrative methods try to purge monetary-policy innovations of the Fed’s systematic response to the economy.
- High-frequency identification avoids much of the conventional endogeneity problem because pre-announcement information is already capitalized into asset prices before the event window.
- The paper’s contribution is to show that even after eliminating standard endogeneity, a second interpretation problem remains: announcements can reveal the Fed’s private or superior information.

Interpretation: The paper changes the interpretation of monetary shocks from “policy surprises” to “announcement surprises.” These surprises can include forward guidance, policy-rule surprises, and signals about fundamentals.

2.3 Contribution relative to New Keynesian evidence on non-neutrality

- A large real-rate response with little inflation response looks like evidence for large nominal and real rigidities under a pure monetary-shock interpretation.
- The paper shows that the same asset-price facts can also arise when FOMC announcements shift beliefs about natural rates and natural output.
- The structural model estimates that about two-thirds of the announcement response is an information effect, while the remaining component is a traditional monetary-policy effect.

Interpretation: This is the paper’s main theoretical contribution: the slope of the Phillips curve and the extent of monetary non-neutrality cannot be inferred from high-frequency real-rate and inflation responses without modeling Fed information effects.

2.4 Contribution relative to central-bank communication

- The paper provides empirical evidence that market participants infer economic information from FOMC announcements.
- It shows that central-bank communication can alter beliefs about fundamentals, not just expectations about future policy instruments.
- This has policy consequences because the information channel can amplify or offset the intended demand channel of policy.

Interpretation: Central-bank announcements are not only tools for changing rates. They are also public signals about the economy.

3 Data and measurement

3.1 FOMC announcement windows

- The monetary shock is measured as the unexpected change in interest rates over a 30-minute window surrounding scheduled FOMC announcements.
- The main sample runs from 2000 to 2014, with 106 regularly scheduled FOMC meetings.
- The narrow window ensures that pre-existing public information is already incorporated into prices before the shock is measured.
- The paper excludes the most acute QE period from July 2008 through June 2009 in the baseline interest-rate regressions.

Interpretation: The narrow event window is the key identification device. It makes the shock plausibly due to the announcement, not to slow-moving macroeconomic confounders.

3.2 Nominal yields, real yields, and breakeven inflation

- Nominal yield responses are measured using Treasury yields and instantaneous forward rates at horizons from three months to ten years.
- Real yield responses use TIPS yields and forwards.
- Breakeven inflation is the difference between nominal and real yields at matching maturities.
- The paper compares nominal, real, and inflation components to assess whether monetary announcements affect real rates or inflation expectations.

Interpretation: Real yields are the key object because monetary non-neutrality requires that monetary policy move real interest rates, not only nominal rates.

3.3 Expected output growth

- The paper uses Blue Chip survey forecasts to measure expected output growth.
- The dependent variable in the output-growth regressions is the revision in expected output growth around FOMC announcements.
- The main result is that expected output growth rises after a surprise increase in interest rates.

Interpretation: This is the decisive evidence for the information effect. A pure contractionary monetary shock should lower expected growth, not raise it.

3.4 Stock prices

- The paper also studies the high-frequency response of stock prices to FOMC announcements.
- Stock prices fall in response to the measured policy-news shock, but the fall is smaller than a model with no information effect would predict.
- This result further supports the idea that a tightening announcement partly conveys good news about fundamentals.

Interpretation: Stock prices react to both higher discount rates and higher expected future cash flows. The information effect raises expected cash flows and dampens the negative discount-rate effect.

4 Empirical specifications

4.1 Baseline high-frequency regression

For an asset return or yield change Δy_t around FOMC announcement t , the baseline regression is:

$$\Delta y_t = \alpha + \beta \cdot Shock_t + \varepsilon_t. \quad (1)$$

- $Shock_t$ is the policy-news shock, scaled so that the one-year Treasury yield rises by 100 basis points.
- Δy_t can be a nominal yield, real yield, breakeven inflation rate, expected output-growth revision, or stock return.
- The coefficient β is the high-frequency response of the dependent variable to the normalized monetary shock.

Interpretation: Scaling by the one-year Treasury yield makes coefficients interpretable as responses to a 100 basis point one-year-yield surprise.

4.2 OLS versus Rigobon-style identification

The paper also checks whether high-frequency estimates are contaminated by nonmonetary shocks using a heteroskedasticity-based identification approach inspired by Rigobon. The idea is that monetary-shock variance is much higher inside the FOMC window than outside it, while background nonmonetary shock variance is similar.

Interpretation: If OLS and Rigobon-style estimates are similar in the 30-minute window, the window is narrow enough that contamination from nonmonetary shocks is limited.

4.3 Output-expectations regression

The output-growth expectation regression is:

$$\Delta E_t[\Delta y_{t+h}] = \alpha_h + \beta_h \cdot Shock_t + u_{t,h}. \quad (2)$$

- $\Delta E_t[\Delta y_{t+h}]$ is the revision in survey expectations about output growth at horizon h .
- The key coefficient is positive for the next-year forecast.

Interpretation: The positive response of expected growth is inconsistent with a pure contractionary policy shock and motivates the structural model.

5 Structural model with Fed information effects

5.1 New Keynesian block

The model is a New Keynesian framework with internal habits and a backward-looking inflation component. Private-sector behavior is summarized by a log-linear Euler equation and Phillips curve:

$$\hat{\lambda}_t^x = E_t \hat{\lambda}_{t+1}^x + (\hat{\iota}_t - E_t \hat{\pi}_{t+1} - \hat{r}_t^n), \quad (3)$$

$$\Delta \hat{\pi}_t = \beta E_t \Delta \hat{\pi}_{t+1} + \kappa \omega \hat{\zeta} \hat{x}_t - \kappa \hat{\zeta} \hat{\lambda}_t^x. \quad (4)$$

- $\hat{\lambda}_t^x$ is the marginal utility gap.
- $\hat{x}_t$ is the output gap.
- $\hat{r}_t^n$ is the natural rate of interest.
- $\kappa \hat{\zeta}$ is the effective slope of the Phillips curve.

Interpretation: The model separates movements in actual real rates from movements in natural rates. This is essential for distinguishing conventional monetary tightening from information about fundamentals.

5.2 Habit formation

With internal habits, the marginal utility gap is:

$$\hat{\lambda}_t^x = -(1 + b^2 \beta) \sigma_c \hat{x}_t + b \sigma_c \hat{x}_{t-1} + b \beta \sigma_c E_t \hat{x}_{t+1}. \quad (5)$$

Interpretation: Habit formation helps the model match the hump-shaped responses of output and interest rates.

5.3 Monetary-policy rule and forward-guidance process

The policy rule is:

$$\hat{l}_t - E_t \hat{\pi}_{t+1} = \bar{r}_t + \phi_\pi \hat{\pi}_t, \quad (6)$$

with

$$\bar{r}_t = (\rho_1 + \rho_2) \bar{r}_{t-1} - \rho_1 \rho_2 \bar{r}_{t-2} + \epsilon_t. \quad (7)$$

Interpretation: The AR(2) process captures the fact that the measured FOMC surprises move expected future interest rates, not just the current short rate.

5.4 Fed information effect

The Fed information effect is modeled as:

$$E_t \hat{r}_{t+j}^n = \psi E_t \bar{r}_{t+j}. \quad (8)$$

The parameter ψ is the fraction of the announcement-induced real-rate movement that is interpreted as news about future natural rates rather than a conventional policy gap.

- If $\psi = 0$, the shock is pure monetary policy.
- If $\psi = 1$, the shock is pure information about fundamentals.
- The baseline estimate is intermediate, around $\psi = 0.68$.

Interpretation: The information effect is the main conceptual innovation in the structural model.

6 Identification and interpretation

6.1 What the high-frequency design identifies

The design identifies the response of asset prices and expectations to information released during FOMC announcements. It does not by itself identify whether that information is pure policy news, forward guidance, macroeconomic news, or a mixture of these.

Interpretation: The paper's message is subtle: the event-window design is credible, but the economic interpretation of the event-window shock is not automatic.

6.2 Why expected output growth matters

If an unexpected rate hike were purely contractionary, expected output growth should fall. Instead, it rises. This means the same announcement that raises rates also reveals good news about economic fundamentals.

Interpretation: The output-expectations fact is the key piece of evidence separating the information-effect interpretation from the conventional monetary-shock interpretation.

6.3 Policy interpretation

A central bank announcement can affect output through two channels:

1. a conventional interest-rate gap channel, where the real rate rises relative to the natural rate and demand falls;
2. an information channel, where private-sector beliefs about future natural output or natural rates change.

Interpretation: The information channel can offset or even reverse the conventional channel in some outcomes.

7 Tables and figures

7.1 Table I: nominal, real, and inflation responses and Figure I: binned scatter for five-year real yield

Read:

- Table I reports the baseline interest-rate responses to the policy-news shock.
- Nominal and real yields respond strongly and persistently; the response of breakeven inflation is small.
- Figure I shows that the positive five-year real-yield response is not driven by a few outliers.

Economic message: The table and figure establish the first major empirical fact: FOMC surprises move real rates far into the term structure.

TABLE I
RESPONSE OF INTEREST RATES AND INFLATION TO THE POLICY NEWS SHOCK

	Nominal	Real	Inflation
3M Treasury yield	0.67 (0.14)		
6M Treasury yield	0.85 (0.11)		
1Y Treasury yield	1.00 (0.14)		
2Y Treasury yield	1.10 (0.33)	1.06 (0.24)	0.04 (0.18)
3Y Treasury yield	1.06 (0.36)	1.02 (0.25)	0.04 (0.17)
5Y Treasury yield	0.73 (0.20)	0.64 (0.15)	0.09 (0.11)
10Y Treasury yield	0.38 (0.17)	0.44 (0.13)	-0.06 (0.08)
2Y Treasury inst. forward rate	1.14 (0.46)	0.99 (0.29)	0.15 (0.23)
3Y Treasury inst. forward rate	0.82 (0.43)	0.88 (0.32)	-0.06 (0.15)
5Y Treasury inst. forward rate	0.26 (0.19)	0.47 (0.17)	-0.21 (0.08)
10Y Treasury inst. forward rate	-0.08 (0.18)	0.12 (0.12)	-0.20 (0.09)

Notes. Each estimate comes from a separate OLS regression. The dependent variable in each regression is the one-day change in the variable stated in the left-most column. The independent variable is a change in the policy news shock over a 30-minute window around the time of FOMC announcements. The sample period is all regularly scheduled FOMC meetings from 1/1/2000 to 3/19/2014, except that we drop July 2008 through June 2009. For two-year and three-year yields and real forwards, the sample starts in January 2004. The sample size for the two-year and three-year yields and forwards is 74. The sample size for all other regressions is 106. In all regressions, the policy news shock is computed from these same 106 observations. Robust standard errors are in parentheses.

Table I: nominal, real, and inflation responses

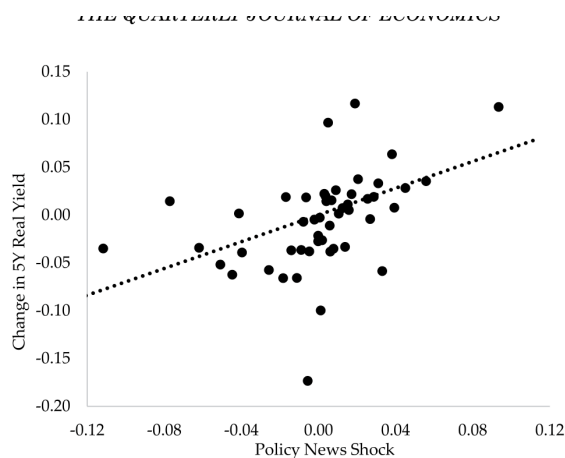


FIGURE I

Binned Scatter Plot for Five-Year Real-Yield Regression

Figure I: binned scatter for five-year real yield

7.2 Table II: OLS versus Rigobon-style estimates and Table III: response of expected output growth

Read:

- Table II compares OLS and heteroskedasticity-based estimates using 30-minute and one-day windows.
- The 30-minute OLS estimates line up closely with the Rigobon-style estimates, suggesting limited contamination.
- Table III shows that expected output growth rises after a surprise rate increase.

Economic message: The shock is cleanly tied to the FOMC window, but it is not a pure contractionary shock because expected output growth rises.

TABLE II
ALLOWING FOR BACKGROUND NOISE IN INTEREST RATES

	2-year forward		5-year forward		10-year forward	
	Nominal	Real	Nominal	Real	Nominal	Real
olicy news shock, 30-minute window						
OLS	1.14 [0.23, 2.04]	0.99 [0.41, 1.57]	0.26 [-0.12, 0.64]	0.47 [0.14, 0.80]	-0.08 [-0.43, 0.28]	0.12 [-0.12, 0.36]
Rigobon	1.10 [0.31, 2.36]	0.96 [0.45, 1.82]	0.22 [-0.14, 0.64]	0.46 [0.15, 0.84]	-0.12 [-0.46, 0.24]	0.11 [-0.13, 0.35]
olicy news shock, 1-day window						
OLS	1.24 [0.80, 1.69]	1.00 [0.57, 1.43]	0.44 [0.18, 0.70]	0.48 [0.20, 0.76]	0.05 [-0.20, 0.29]	0.15 [-0.10, 0.39]
Rigobon	0.93 [-0.64, 2.08]	0.82 [0.38, 3.20]	-0.11 [-1.23, 0.33]	0.33 [-0.07, 1.12]	-0.51 [-1.93, -0.08]	-0.04 [-0.51, 0.45]
year nominal yield, 1-day window						
OLS	1.23 [1.07, 1.38]	0.94 [0.69, 1.20]	0.64 [0.43, 0.84]	0.54 [0.31, 0.76]	0.18 [0.01, 0.35]	0.20 [0.02, 0.38]
Rigobon (90% CI)	1.14 [0.82, 1.82]	0.82 [0.62, 2.98]	-0.11 [-7.94, 0.60]	0.33 [-0.01, 7.48]	-0.51 [-10.00, -0.21]	-0.04 [-4.57, 0.38]

Notes. Each estimate comes from a separate "regression." The dependent variable in each regression is the one-day change in the variable stated at the top of that column. In the first panel of results is the 30-minute change in the policy news shock around FOMC meeting times, in the second panel it is the one-day change in the policy news shock, and in the third panel it is the one-day change in the two-year nominal yield. In each panel, we report results based on OLS and Rigobon's heteroskedasticity-based estimation approach. We report a point estimate and 95% confidence intervals except in the last row of Rigobon estimates, which reports 90% confidence intervals. The sample of "treatment" days for the Rigobon method is all regularly scheduled FOMC meeting days from 1/1/2000 to 3/19/2014; this is also the period for which the key news shock is constructed in all "regressions." The sample of "control" days for the Rigobon analysis is all Tuesdays and Wednesdays that are not FOMC meeting days over a same period of time. In both the treatment and control samples, we drop July 2008 through June 2009 and 8/1/2001-8/2/2001. For two-year forwards, the sample starts in 1995m1-2014m4.

Table II: OLS versus Rigobon-style estimates

TABLE III RESPONSE OF EXPECTED OUTPUT GROWTH OVER THE NEXT YEAR				
	1995–2014	2000–2014	2000–2007	1995–2000
Policy news shock	1.01 (0.32)	1.04 (0.35)	0.95 (0.32)	0.79 (0.63)
Observations	120	90	52	30

Notes. We regress changes from one month to the next in survey expectations about output growth over the next year from the Blue Chip Economic Indicators on the policy news shock that occurs in that month (except that we drop policy news shocks that occur in the first week of the month because we do not know whether these occurred before or after the survey response). Specifically, the dependent variable is the change in the average forecasted value of output growth over the next three quarters (the maximum horizon over which forecasts are available for the full sample). See Online Appendix F for details. We present results for four sample periods. The longest sample period we have data for is 1995m1–2014m4; this is also the period for which the policy news shocks are constructed. We also present results for 2000m1–2014m4 (which corresponds

Table III: response of expected output growth

7.3 Figure II: binned scatter for expected output growth and Table IV: structural parameter estimates

Read:

- Figure II shows the positive relationship between policy news and output-growth expectations.
- Table IV reports the structural estimates. The baseline information parameter is $\psi = 0.68$.
- The estimated Phillips-curve slope is very small, but ignoring information effects changes the interpretation of the slope.

Economic message: The structural model reconciles high real-rate responses with rising growth expectations by allowing FOMC announcements to convey news about fundamentals.

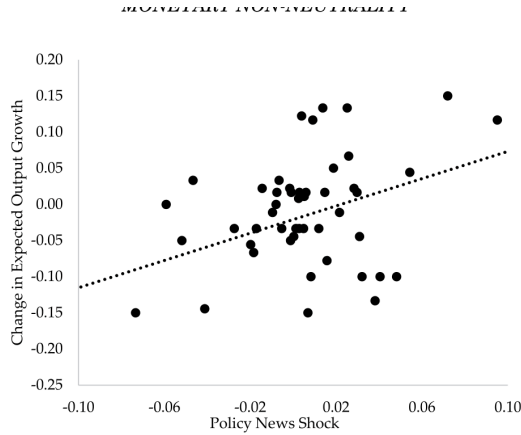


FIGURE II

Binned Scatter Plot for Expected Output Growth Regression

Figure II: binned scatter for expected output growth

TABLE IV ESTIMATES OF STRUCTURAL PARAMETERS				
	ψ	$\kappa \zeta \times 10^{-5}$	ρ_1	ρ_2
Baseline	0.68 [0.33, 0.84]	11.2 [0.0, 60.2]	0.9 [0.83, 0.96]	0.79 [-0.69, 0.89]
No Fed information ($\psi = 0$)	0.00 —	3.4 [0.0, 24.1]	0.9 [0.83, 0.96]	0.79 [-0.69, 0.89]
Full Fed information ($\psi = 0.99$)	0.99 —	563 [0, 12,538]	0.9 [0.82, 0.96]	0.79 [-0.67, 0.89]
Lower IES ($\sigma = 0.25$)	0.67 [0.25, 0.89]	13.7 [0.0, 94.6]	0.9 [0.83, 0.96]	0.79 [-0.69, 0.89]
Higher IES ($\sigma = 1$)	0.68 [0.42, 0.81]	8.2 [0.0, 44.0]	0.9 [0.83, 0.96]	0.79 [-0.69, 0.89]
No habits ($b = 0$)	1 [0.92, 1.00]	1,000 [0, 43,236]	0.9 [0.83, 0.96]	0.79 [-0.69, 0.89]

Notes. The table reports the estimates of the structural parameters of the model estimated in the

Table IV: structural parameter estimates

7.4 Figure III: rates and inflation and Figure IV: expected output growth and output gap, Figure V: real-rate decomposition and Figure VI: actual and natural output

Read:

- Figure III shows the model response of nominal rates, real rates, and inflation to a contractionary announcement.
- Figure IV shows that expected output growth rises while the output gap falls.

- Figure V decomposes the real-rate response into the natural-rate component and the conventional interest-rate-gap component.
- Figure VI shows the response of actual output, natural output, and the output gap.

Economic message: The model produces a hybrid shock: the announcement raises fundamentals, but policy rates rise by more than the natural rate, so the output gap contracts.

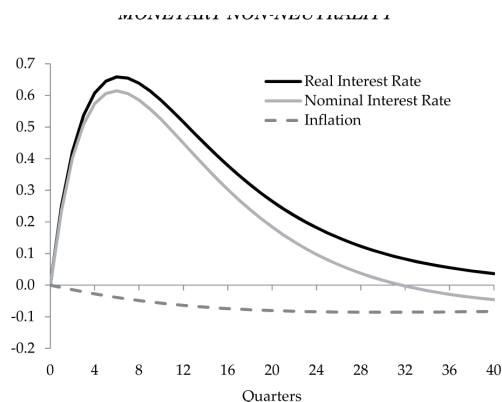


FIGURE III

Responses of Nominal and Real Rates and Inflation to a Contractionary Shock

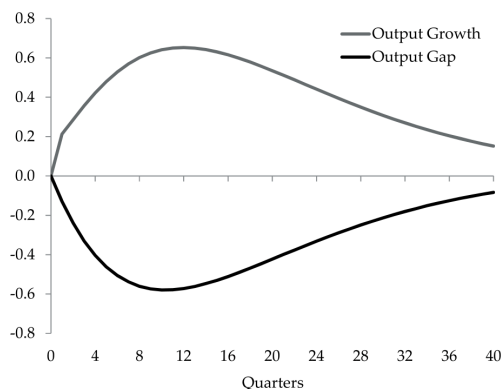


FIGURE IV

Responses of Expected Output Growth and Output Gap to a Contractionary Shock

4 *Figure III: rates and inflation*
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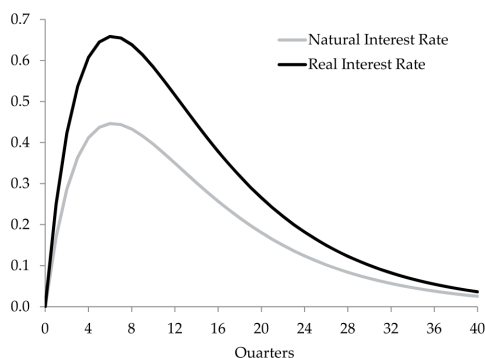


FIGURE V

decomposition of Real Rate into Interest Rate Gap and Natural Interest Rate
Figure V: real-rate decomposition

and Figures III–V illustrate the fit of the model. As in the data, *Figure IV: expected output growth and output gap*
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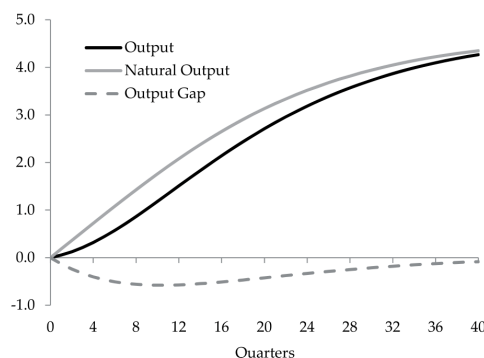


FIGURE VI

Response of Actual and Natural Expected Output
Figure VI: actual and natural output

7.5 Figure VII: actual versus counterfactual output, Figure VIII: causal effect on expected output, Figure IX: expected growth with and without information and Table V: response of stock prices

Read:

- Figure VII compares actual expected output with a counterfactual that removes the Fed information effect.
- Figure VIII plots the causal effect of monetary shocks on expected output, decomposing the output and natural-output channels.
- Figure IX contrasts expected output growth with and without information effects.
- Table V shows that stock prices fall less than in a no-information-effect model.

Economic message: The information channel substantially changes the causal interpretation of monetary announcements and helps explain the stock-market response.

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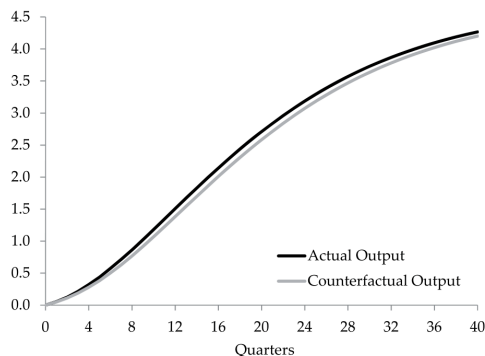


FIGURE VII

Response of Actual and Counterfactual Expected Output
Figure VII: actual versus counterfactual output

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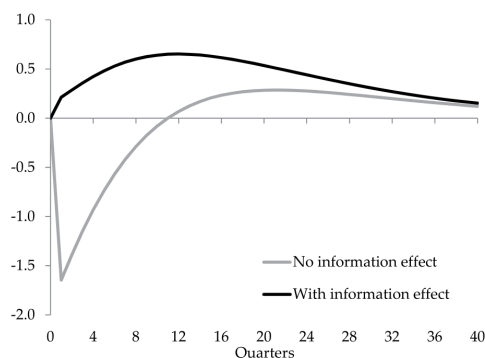


FIGURE IX

Expected Output Growth with and without the Information Effect
Figure IX: expected growth with and without information

MONETARY POLICY TABLE 1

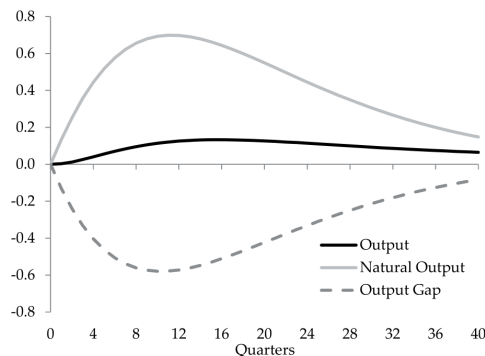


FIGURE VIII

Causal Effect of Monetary Shocks on Expected Output
Figure VIII: causal effect on expected output

TABLE V
RESPONSE OF STOCK PRICES

	Stock prices
Response in the data	-6.5 (3.9)
Response in the model	
Baseline	-6.8 [-11.5, -1.6]
No Fed Information effect	-11.1 [-19.4, -2.5]

Table V: response of stock prices

7.6 Appendix robustness tables

Read:

- Appendix Table A.1 reports additional nominal and real interest-rate responses.
- Appendix Table A.2 repeats the analysis using unexpected changes in federal funds futures.
- Appendix Table A.3 gives additional Rigobon-style estimates.
- Appendix Tables A.4 and A.5 report breakeven inflation and output-growth expectation robustness.

Economic message: The appendix tables show that the central facts are robust across alternative samples, shock measures, and expectation horizons.

RESPONSE OF INTEREST RATES TO MONETARY SHOCKS FOR DIFFERENT SAMPLE PERIODS

	Baseline sample		Precrisis (2000–2007)		Full sample		Baseline w/ unsched.	
	Nominal	Real	Nominal	Real	Nominal	Real	Nominal	Real
3M Treasury yield	0.67 (0.14)		0.78 (0.13)		0.73 (0.15)		0.76 (0.11)	
6M Treasury yield	0.85 (0.11)		0.85 (0.12)		0.90 (0.12)		0.91 (0.10)	
1Y Treasury yield	1.00 (0.14)		1.00 (0.14)		1.00 (0.13)		1.00 (0.13)	
2Y Treasury yield	1.10 (0.33)	1.06 (0.24)	1.11 (0.26)	1.04 (0.24)	1.19 (0.29)	1.46 (0.39)	1.30 (0.27)	1.21 (0.26)
3Y Treasury yield	1.06 (0.36)	1.02 (0.25)	1.05 (0.26)	0.97 (0.25)	1.21 (0.31)	1.34 (0.33)	1.26 (0.40)	1.18 (0.28)
5Y Treasury yield	0.73 (0.20)	0.64 (0.15)	0.67 (0.20)	0.58 (0.15)	0.78 (0.18)	0.81 (0.16)	0.69 (0.16)	0.68 (0.11)
10Y Treasury yield	0.28 (0.17)	0.44 (0.13)	0.26 (0.18)	0.44 (0.13)	0.20 (0.17)	0.57 (0.16)	0.28 (0.13)	0.43 (0.10)
2Y Tr. inst. forward rate	1.14 (0.46)	0.99 (0.29)	1.07 (0.40)	0.90 (0.27)	1.31 (0.38)	0.97 (0.36)	1.34 (0.40)	1.15 (0.31)
3Y Tr. inst. forward rate	0.82 (0.43)	0.88 (0.32)	0.68 (0.43)	0.76 (0.29)	1.14 (0.41)	1.09 (0.35)	1.00 (0.40)	1.03 (0.34)
5Y Tr. inst. forward rate	0.26 (0.19)	0.47 (0.17)	0.29 (0.19)	0.47 (0.16)	0.44 (0.21)	0.61 (0.21)	0.27 (0.13)	0.45 (0.12)
10Y Tr. inst. forward rate	-0.08 (0.18)	0.12 (0.12)	-0.01 (0.19)	0.21 (0.13)	0.05 (0.17)	0.10 (0.13)	-0.07 (0.12)	0.04 (0.11)

Note: Each estimate comes from a separate OLS regression. The dependent variable in each regression is the one-day change in the variable stated in the first left-most column. The independent variable is a change in the policy rate over a 30-day window around quarterly scheduled FOMC announcements, except the last two columns where we include the unscheduled component of the policy rate. The sample period for the baseline is 1954:1–1979:4, for the precrisis sample is 2000:1–2007:4, and for the full sample is 1954:1–2007:4. The sample size for the baseline is 112, for the precrisis sample is 84, and for the full sample is 200. The “full sample” is 1/1950:1 to 1/1950:4 in the first two columns, we exclude a 30-day period after 9/11/2001. For two-year and three-year yields and real rates, the sample starts at 1959:1. The standard errors are in parentheses.

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Notes: Each estimate comes from a separate OLS regression. The dependent variable in each regression is the one-day change in the variable stated in the variable stated in the left-most column. The independent variable is a change in the policy news shock over a 30-minute window around regularly scheduled FOMC announcements, except the last two columns where we include unscheduled FOMC announcements. The baseline sample period is 1/2000 to 3/19/2014, except that we drop July 2008 through June 2009. The "precisus" sample is January 2000 through December 2007. The "full sample" is 1/2000 to 3/19/2014. In the last two columns, we estimate a 10-day period after 9/11/2001. For two-year and three-year yields and real forwards, the sample starts in 2004. For each sample period, we construct the policy news shocks from the same sample of observations as the regressions are run on, that is, the results from the different sample periods use slightly different policy news shock series. Robust standard errors are in parentheses.

Appendix Table A.1

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APPENDIX TABLE A.3 RESPONSES TO POLICY NEWS SHOCK USING THE RIGOBON ESTIMATOR			
	Nominal	Real	Inflation
3M Treasury yield	0.69 (0.15)		
6M Treasury yield	0.85 (0.12)		
1Y Treasury yield	0.98 (0.15)		
2Y Treasury yield	1.07 (0.37)	1.03 (0.29)	0.05 (0.20)
3Y Treasury yield	1.03 (0.42)	0.99 (0.30)	0.04 (0.19)
5Y Treasury yield	0.69 (0.22)	0.62 (0.16)	0.07 (0.12)
10Y Treasury yield	0.34 (0.19)	0.42 (0.14)	-0.08 (0.09)
2Y Treasury inst. forward rate	1.10 (0.51)	0.96 (0.34)	0.14 (0.25)
3Y Treasury inst. forward rate	0.78 (0.49)	0.86 (0.37)	-0.08 (0.17)
5Y Treasury inst. forward rate	0.22 (0.20)	0.46 (0.18)	-0.24 (0.09)
10Y Treasury inst. forward rate	-0.12 (0.19)	0.11 (0.13)	-0.22 (0.10)

Notes: Each estimate comes from a separate "regression." The dependent variable in each regression is the one-day change in the variable stated in the left-most column. The independent variable is a change in the policy news shock over a 30-minute window around the time of FOMC announcements. All results are based on Rigobon's (2003) method of identification by heteroskedasticity. The sample of "treatment" days for the Rigobon method is all regularly scheduled FOMC meeting days from 1/1/2000 to 3/19/2014; this is also the period for which the policy news shock is constructed in all regressions. The sample of control days for the Rigobon analysis is all Tuesdays and Wednesdays that are not FOMC meeting days over the same period of time. In both the treatment and control samples, we drop July 2008 through June 2009 and 9/11/2001–9/21/2001. For two-year forwards, the sample starts in January 2004. Standard errors are calculated using a nonparametric bootstrap with 5,000 iterations.

Appendix Table A.3

8 Short summary

- The paper measures monetary shocks using 30-minute windows around scheduled FOMC announcements.
- A surprise rate hike raises nominal and real yields several years into the term structure.
- Inflation expectations barely move at short horizons and only modestly decline at longer horizons.
- Expected output growth rises after surprise rate hikes.

APPENDIX TABLE A.2
RESPONSE TO A FED FUNDS RATE SHOCK

	Nominal	Real	Inflation
3M Treasury yield	0.50 (0.16)		
6M Treasury yield	0.59 (0.10)		
1Y Treasury yield	0.41 (0.16)		
2Y Treasury yield	0.48 (0.32)	0.50 (0.20)	-0.02 (0.15)
3Y Treasury yield	0.38 (0.34)	0.41 (0.19)	-0.03 (0.18)
5Y Treasury yield	0.11 (0.16)	0.21 (0.12)	-0.10 (0.09)
10Y Treasury yield	-0.00 (0.12)	0.10 (0.09)	-0.10 (0.07)
2Y Treasury inst. forward rate	0.29 (0.40)	0.30 (0.20)	-0.01 (0.25)
3Y Treasury inst. forward rate	0.07 (0.34)	0.13 (0.19)	-0.06 (0.22)
5Y Treasury inst. forward rate	-0.09 (0.13)	0.07 (0.12)	-0.16 (0.07)
10Y Treasury inst. forward rate	-0.11 (0.16)	-0.02 (0.11)	-0.08 (0.09)

Notes: Each estimate comes from a separate OLS regression. The dependent variable in each regression is the one-day change in the variable stated in the left-most column. The independent variable is a change in the Fed funds future over the remainder of the month over a 30-minute window around the time of FOMC announcements. The sample period is all regularly scheduled meetings from 1/1/2000 to 3/19/2014, except that we drop July 2008 through June 2009. For two-year and three-year yields and real forwards, the sample starts in January 2004. The sample size for the two-year and three-year yields and forwards is 74. The sample size for all other regressions is 106. In all regressions, the policy news shock is computed from these same 106 observations. Robust standard errors are in parentheses.

Appendix Table A.2

MONETARY POLICY TRANSMISSION

APPENDIX TABLE A.4
BREAK-EVEN INFLATION VERSUS INFLATION SWAPS

	Breakeven	Swaps
Inflation over next 2 years	-0.02 (0.18)	0.37 (0.35)
Inflation over next 3 years	-0.03 (0.17)	0.41 (0.32)
Inflation over next 5 years	-0.13 (0.14)	-0.02 (0.15)
Inflation over next 10 years	-0.22 (0.12)	-0.17 (0.16)

Notes: Each estimate comes from a separate OLS regression. The dependent variable in each regression is the one-day change in expected inflation measured either by break-even inflation from the difference between nominal Treasuries and TIPS (first column) or from inflation swaps (second column) for the period stated in the left-most column. The independent variable is a change in the policy news shock over a 30-minute window around the time of FOMC announcements, where the policy news shocks are constructed on the 2000–2014 sample used in Table 1. The sample is all regularly scheduled FOMC meeting days from 1/1/2000 to 11/14/2012, except that we drop July 2008 through June 2009. Robust standard errors are in parentheses.

APPENDIX TABLE A.5
RESPONSE OF EXPECTED OUTPUT GROWTH

Exp. output growth in current qr	1.38 (0.72)
Exp. output growth 1 qr ahead	1.56 (0.54)
Exp. output growth 2 qr ahead	0.66 (0.37)
Exp. output growth 3 qr ahead	0.82 (0.27)
Exp. output growth 4 qr ahead	0.51 (0.25)
Exp. output growth 5 qr ahead	0.54 (0.22)

Appendix Tables A.4 and A.5

- This output-expectations result is inconsistent with a pure contractionary monetary shock.
- The authors propose that FOMC announcements contain Fed information about fundamentals.
- A New Keynesian model with Fed information effects estimates that roughly two-thirds of the announcement shock is information about natural rates/fundamentals.
- Ignoring information effects can overstate monetary non-neutrality or misinterpret the Phillips-curve slope.
- The paper’s main lesson is that high-frequency monetary shocks are clean announcement shocks, but not necessarily pure policy shocks.

9 Conclusion

9.1 Core conclusion

Nakamura and Steinsson show that high-frequency monetary surprises around FOMC announcements move real rates strongly but also raise expected output growth. This combination is hard to reconcile with a pure contractionary monetary-policy shock. The evidence implies that FOMC announcements reveal information about fundamentals.

9.2 Economic intuition

When the Fed unexpectedly raises rates, markets may infer that the Fed is more optimistic about future economic fundamentals. This good-news component raises expected output growth and partially offsets the usual contractionary effect of higher real rates. The observed shock is therefore a mixture of a policy component and an information component.

9.3 Takeaway

High-frequency identification remains powerful, but researchers must interpret the identified event-window surprises carefully. The paper’s key contribution is to show that central-bank announcements are signals, not just policy actions. Monetary non-neutrality should therefore be inferred from models that explicitly account for the information content of policy announcements.